ALICE L WALTON SCHOOL OF MEDICINE

Retroviridae (Part 1 of 2)

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Editor: Jordan Finn

Adapted from: Retroviridae

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Differentiate between the taxonomy of Retroviridae and other viral families.
2. Explain disease progression and epidemiology of HIV.
3. Describe the life cycle of HIV.

Retroviridae: Taxonomy (LO1)

Retroviridae include two medically significant viruses: Human immunodeficiency virus (HIV) and Human T-cell lymphotropic virus type 1 (HTLV-1). HIV is the virus that causes acquired immunodeficiency syndrome (AIDS). HTLV-1 causes T cell cancers, most commonly in immunosuppressed patients.

Retroviruses encode their genetic information as two copies of ssRNA, but use an enzyme called reverse transcriptase to convert viral RNA into DNA. This DNA viral copy is then incorporated into host cells and remains there. While most other RNA viruses replicate in the cytoplasm of host cells, retroviruses replicate in the nucleus once their DNA has been incorporated into the host cell's genome.

All retroviruses have 3 genes: *env, gag,* and *pol*. HIV contains 6 additional genes (9 genes total) that encode proteins that allow the virus to evade and suppress the host immune system.

Retroviridae viruses also have an envelope. This envelope is derived from the host plasma membrane.

What are the characteristics of Retroviridae members?

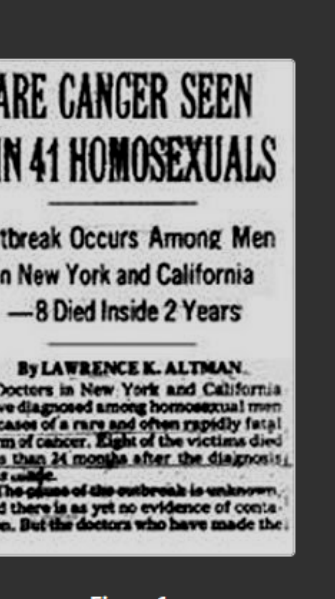
When was HIV first identified in the U.S.? (LO2)

In 1981, clusters of previously healthy young men were diagnosed with Kaposi sarcoma, a rare opportunistic cancer, and *Pneumocystis jirovecii*, an opportunistic respiratory fungal infection (Figure 1). Kaposi sarcoma is caused by a herpesvirus, **HHV-8** (Figure 2). These conditions were typically seen only in individuals with severe immunosuppression or advanced age, making their appearance in young adults highly unusual. This outbreak initially centered around Los Angeles, San Francisco, and New York.

CLINICAL CORRELATION

HHV-8 (KSHV) is a herpesvirus that mainly causes problems in immunocompromised people, especially those with HIV. It most commonly leads to Kaposi sarcoma (KS), a vascular tumor that arises from endothelial cells. KS appears as red-purple spots or nodules on the skin or in the mouth, often starting on the legs or face and sometimes affecting internal organs.

The virus stays latent in B cells and endothelial cells; transmission occurs via saliva, sexual conta



RARE CANCER SEEN IN 41 HOMOSEXUALS

Outbreak Occurs Among Men in New York and California —8 Died Inside 2 Years

By LAWRENCE K. ALTMAN

Doctors in New York and California have diagnosed among homosexual men 41 cases of a rare and often rapidly fatal form of cancer. Eight of the victims died less than 24 months after the diagnosis was made.

The cause of the outbreak is unknown, and there is as yet no evidence of contagion. But the doctors who have made the

Figure 1

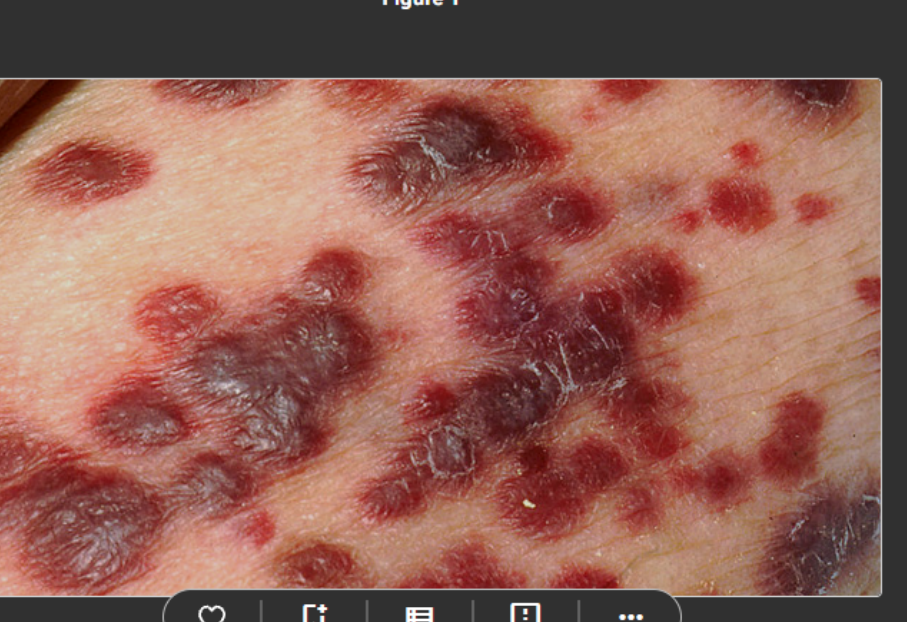


Figure 2: Skin lesions caused by Kaposi sarcoma.

Upon further investigation, all of these young men were suffering from a severely deficient lymphocyte count, with CD4 T cell counts being crushingly low. Because the earliest clusters were predominantly among men who have sex with me, this mysterious illness was originally called Gay-Related Immune Deficiency. The naming was not only inaccurate, as the disease would be rapidly identified in other demographics, but also fueled rampant misinformation and stigma that persists to this day.

As the disease continued to spread to other demographics it was renamed **acquired immunodeficiency syndrome (AIDS)**. People who inject drugs, hemophiliacs receiving blood products, children birthed by infected people, and heterosexual partners of infected individuals were all at risk. This pattern led to the understanding that the disease is transmitted through blood, sexual contact, and vertically. It would take years to discover the cause of AIDS after the initial outbreak in the U.S. occurred.

By 1983, researchers led by Luc Montagnier and Françoise Barré-Sinoussi discovered the human immunodeficiency virus (HIV). Widespread screening tests for HIV were not available until 1985, which meant the virus had already spread extensively before routine testing and preventive measures could be implemented.

Wait, there's more than one HIV? (LO2)

HIV is categorized into types and subtypes. There are two types of HIV: HIV-1 and HIV-2.

- **HIV-1.** The most common and widespread type of HIV and is responsible for the majority of infections globally.
- **HIV-2.** Less common and transmissible than HIV-1, HIV-2 is mostly found in West Africa. HIV-2 infection also progresses more slowly to AIDS, making it more manageable.

Both HIV-1 and HIV-2 are believed to have originated in Central Africa, but were transmitted independently to humans in separate incidences. HIV-1 is believed to originated in chimpanzees while HIV-2 originated in sooty mangabey monkeys. The most probable route of transmission was through hunting and consumption of bushmeat, which exposed humans to infected blood and tissues.

Starting around the 1920s, HIV-1 began spreading outward from Kinshasa, the largest city in what is now the Democratic Republic of the Congo. Factors such as rapid urbanization, the sex trade, and increased population density contributed to the virus's adaptation and transmission among humans. By the 1960s, HIV had reached Haiti, and although HIV/AIDS was first recognized in the United States in the early 1980s, evidence suggests the virus had been present in the country for at least a decade prior.

Which type of HIV is responsible for most cases of HIV globally?

How does HIV spread and who is at risk? (LO2)

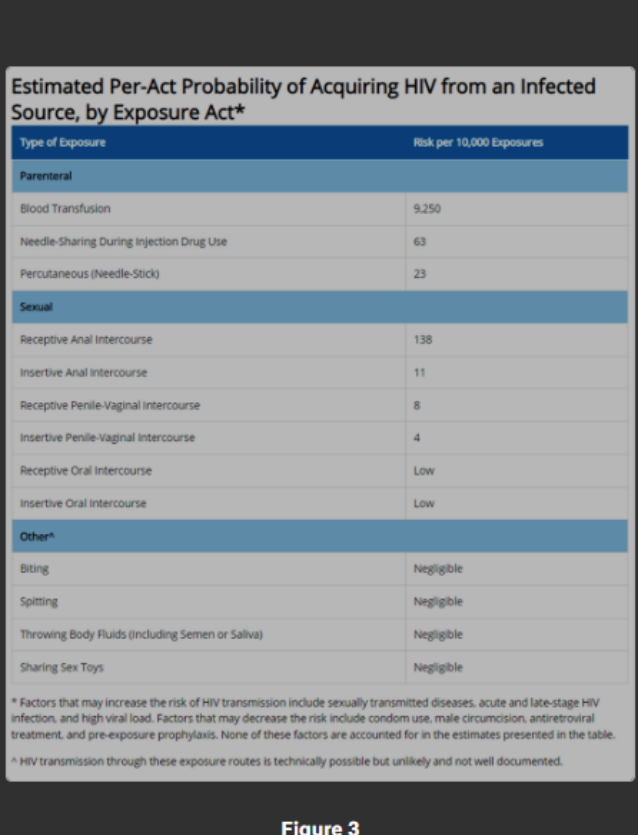
HIV is a bloodborne pathogen. As such, casual contract and respiratory secretions do not transmit the virus. Here are some of the main ways the virus is transmitted:

- **Sexual:** HIV can be transmitted through unprotected sexual contact. Certain practices, such as anal intercourse, carry a higher risk due to mucosal vulnerability.
- **Parenteral:** Exposure to infected blood or blood products, including sharing needles or accidental needle sticks. Before routine screening, blood transfusions and organ transplants were common sources of transmission.
- **Occupational:** Healthcare workers may be exposed to blood or body fluids during procedures if proper precautions are not taken.
- **Vertical Transmission:** HIV is a TORCH pathogen and can cross the placenta during pregnancy. It can also be transmitted during delivery or through breastfeeding.

CLINICAL CORRELATION

Without antiretroviral therapy (ART), the risk of transmitting HIV to an infant is 1 in 3. This can either oc feeding. If a pregnant person is on ART, the risk of transmission is near zero. Screening for HIV and access to treatment are critical to stop the spread of HIV.

Exposure to HIV does not necessarily mean transmission will occur. Here's a summary of the rate of transmission per 10,000 exposures (Figure 3).



Estimated Per-Act Probability of Acquiring HIV from an Infected Source, by Exposure Act*

Type of Exposure	Risk per 10,000 Exposures
Parenteral	
Blood Transfusion	9,250
Needle-Sharing During Injection Drug Use	63
Percutaneous (Needle-Stick)	23
Sexual	
Receptive Anal Intercourse	138
Insertive Anal Intercourse	11
Receptive Penile-Vaginal Intercourse	8
Insertive Penile-Vaginal Intercourse	4
Receptive Oral Intercourse	Low
Insertive Oral Intercourse	Low
Other^	
Biting	Negligible
Spitting	Negligible
Throwing Body Fluids (including Semen or Saliva)	Negligible
Sharing Sex Toys	Negligible

* Factors that may increase the risk of HIV transmission include sexually transmitted diseases, acute and late-stage HIV infection, and high viral load. Factors that may decrease the risk include condom use, male circumcision, antiretroviral treatment, and pre-exposure prophylaxis. None of these factors are accounted for in the estimates presented in the table.

^ HIV transmission through these exposure routes is technically possible but unlikely and not well documented.

Figure 3

Credit: CDC

Blood transfusions, n se, and receptive anal intercourse have the highest risk of transmission. Blood transfusions are now screened for HIV and other bloodborne pathogens.

Simple harm-reduction strategies can significantly reduce the risk of HIV transmission. For example, people who inject drugs should avoid sharing needles, as this practice can spread both HIV as well as other bloodborne infections. Reusing needles can also cause physical damage because they become dull over time (Figure 4). Some individuals attempt to lick needles to ease injection, but this, combined with needle sharing, greatly increases the risk of infection. Access to clean needles and education on safe practices are essential for prevention.

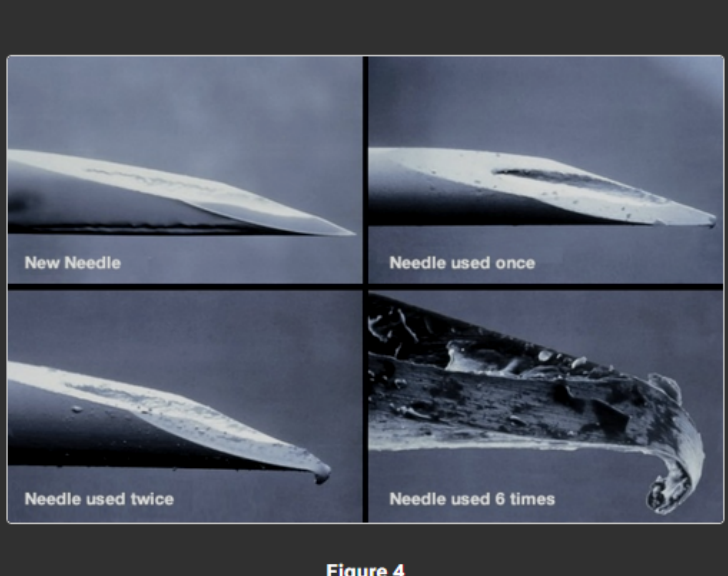


Figure 4

Which actions have the highest risk of transmitting HIV per exposure?

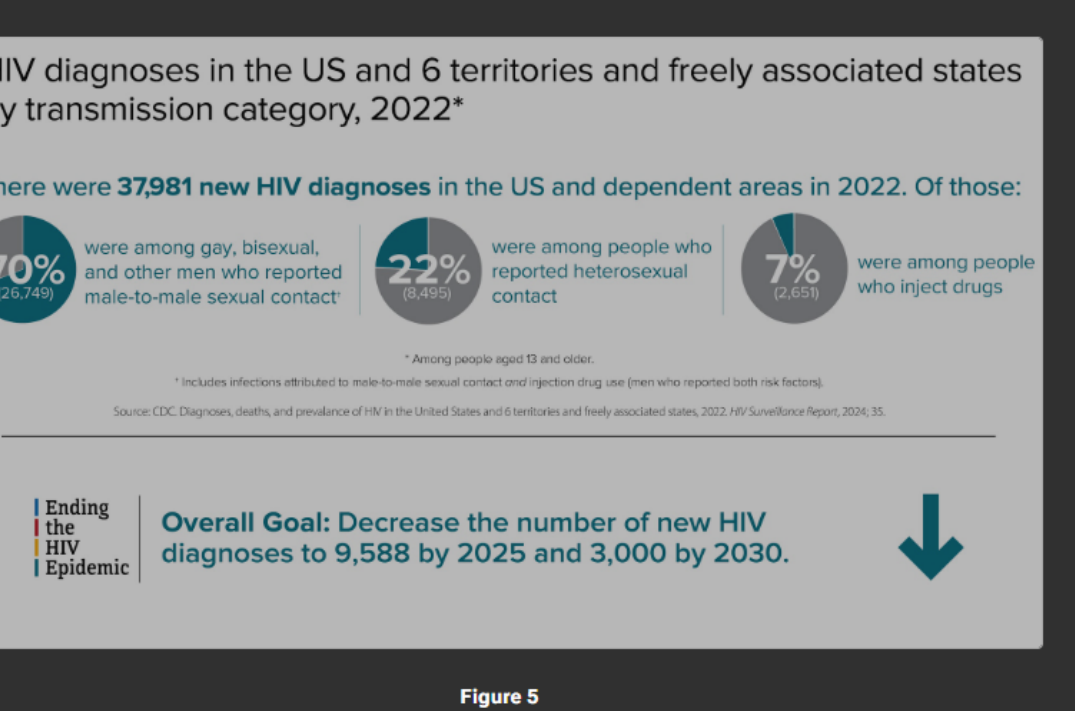
Sexual contact is a common route of HIV transmission, but using protection such as condoms is highly effective in reducing risk. For individuals with HIV infection, consistent use of antiretroviral therapy (ART) can lower viral levels in the blood to undetectable levels. With prophylactic therapies and testing, ART can dramatically reduce transmission, as higher viral loads are strongly associated with increased risk.

CLINICAL CORRELATION

Pre-exposure prophylaxis (PrEP) is a medication that helps prevent HIV infection in people at high risk. When taken consistently as prescribed, PrEP reduces the risk of acquiring HIV by more than 90%. Ideally, abstaining from risky behavior would be the best option but if not possible, PrEP is an excellent alternative.

Regular testing, knowing one's HIV status, and ensuring that individuals living with HIV start and stay on antiretroviral therapy are key strategies to minimize transmission. These measures protect the health of the individual, their partners, and future children.

In the U.S. HIV disproportionately affects several demographics (Figure 5) (Figure 6).



HIV diagnoses in the US and 6 territories and freely associated states by transmission category, 2022*

There were **37,981 new HIV diagnoses** in the US and dependent areas in 2022. Of those:

- 70% (26,749) were among gay, bisexual, and other men who reported male-to-male sexual contact†
- 22% (8,495) were among people who reported heterosexual contact
- 7% (2,651) were among people who inject drugs

* Among people aged 13 and older.

† Includes infections attributed to male-to-male sexual contact and injection drug use (men who reported both risk factors).

Source: CDC. Diagnoses, deaths, and prevalence of HIV in the United States and 6 territories and freely associated states, 2022. HIV Surveillance Report, 2024; 35.

Ending the HIV Epidemic

Overall Goal: Decrease the number of new HIV diagnoses to 9,588 by 2025 and 3,000 by 2030.

Figure 5

Credit: CDC

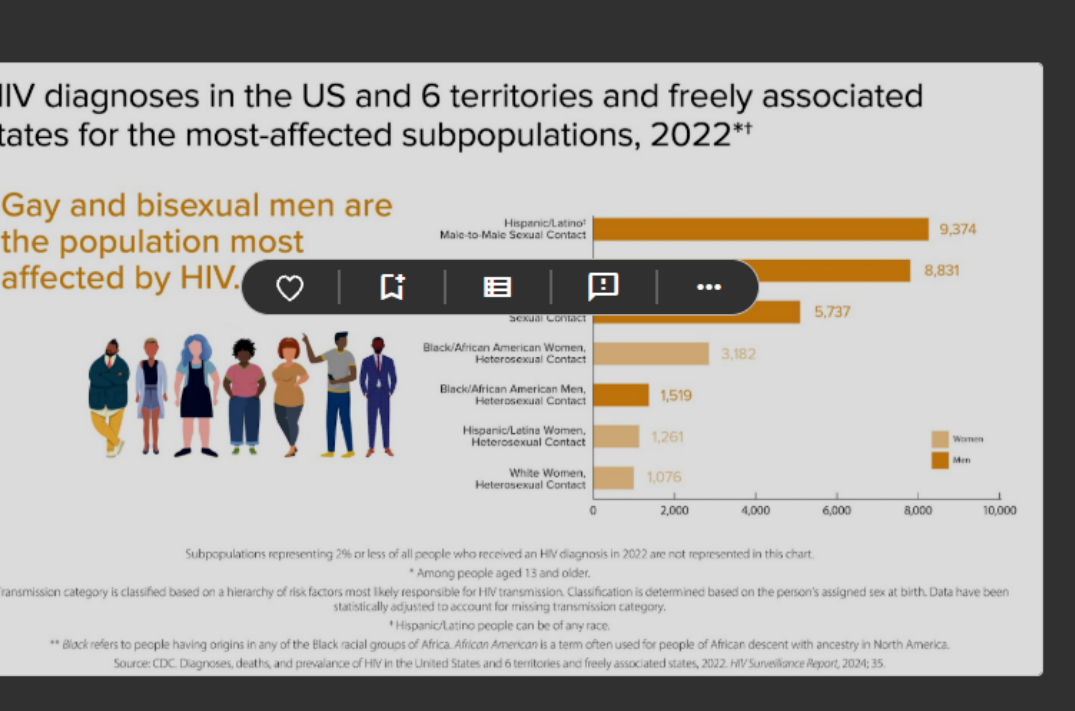


Figure 6

Among all demographics, Black and Hispanic/Latino men who have sex with men (MSM) experience the highest burden of HIV in the United States. However, it's important to note that approximately one in four new HIV diagnoses occur in heterosexual individuals. While HIV disproportionately affects certain populations, it remains a public health issue for everyone.

How are efforts to control HIV globally going? (LO2)

HIV is one of the most devastating pathogens in the history of humanity. Since the beginning of the pandemic, 88.4 million people worldwide have become infected with HIV, 42.3 million people have died. In 2023, 39.9 million people globally are living with HIV, with 1.3 million new cases in 2023.

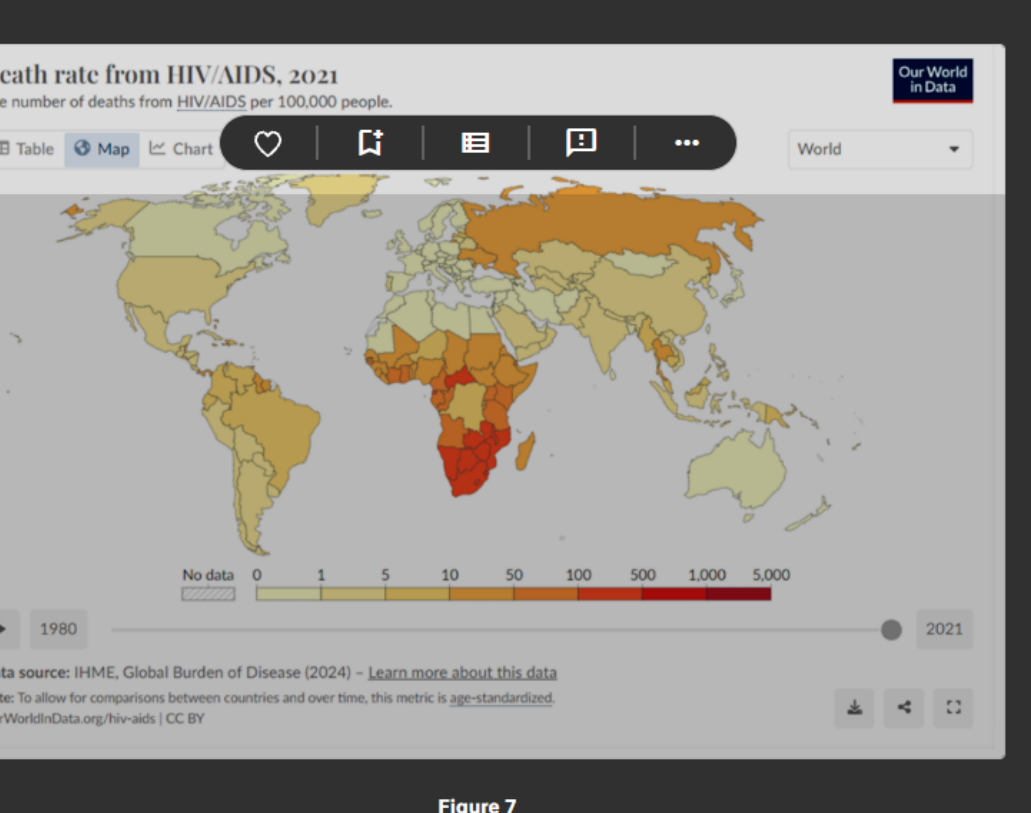

Death rate from HIV/AIDS, 2021
The number of deaths from HIV/AIDS per 100,000 people.
Our World in Data
Table
Map
Chart
World
No data
0
1
5
10
50
100
500
1,000
5,000
1980
2021
Data source: IHME, Global Burden of Disease (2024) – Learn more about this data
Note: To allow for comparisons between countries and over time, this metric is age-standardized.
OurWorldInData.org/hiv-aids | CC BY

Figure 7

Credit: https://ourworldindata.org/hiv-aids

The UN's goal is to end AIDS as a threat by 2030. While significant progress has been made, there are still challenges. Increasing access to ART, PrEP, and HIV education have drastically reduced the number of new cases and deaths due to HIV globally (Figure 8).

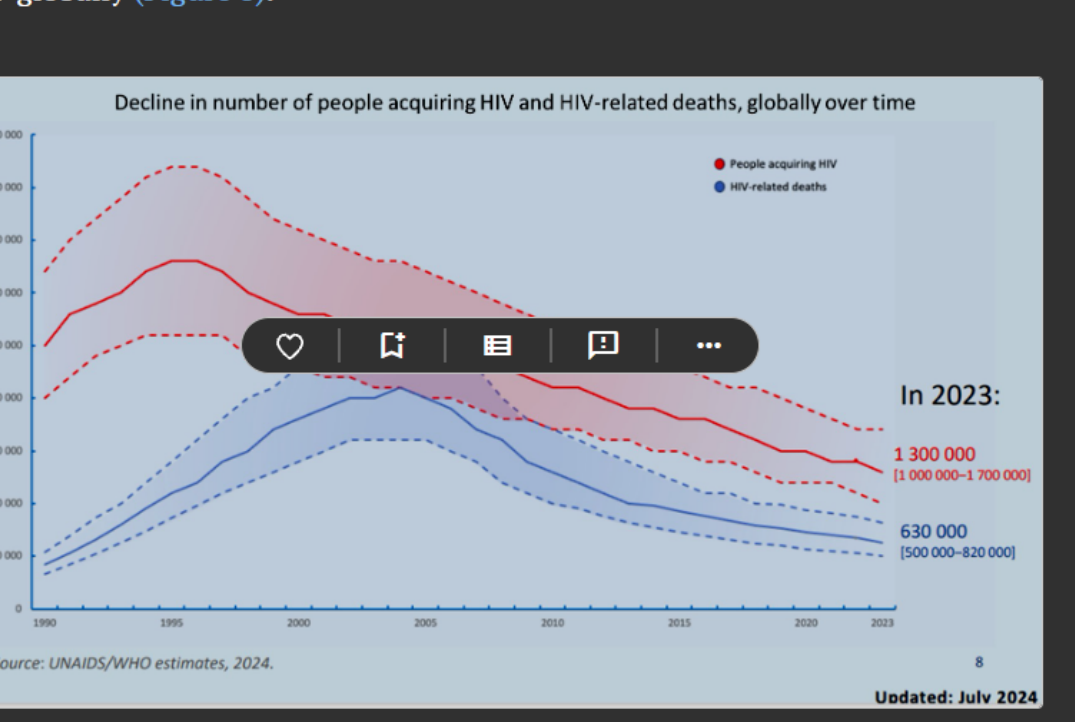


Figure 8

However, in impoverished areas of the world where access to healthcare is poor, rates of HIV infection continue to be high. Sub-Saharan Africa has an incredibly high rate of HIV infections as well as co-infections with *M. tuberculosis*. The immunosuppression caused by HIV facilitates the reactivation and spread of tuberculosis and other diseases as well.

How are efforts to control HIV in the U.S. going? (LO2)

As with the global data, progress has been made. However, there are regions of the U.S. disproportionately affected by HIV. Remember that HIV was first recognized in San Francisco and New York. Where do you think HIV rates are the highest in the U.S. today?

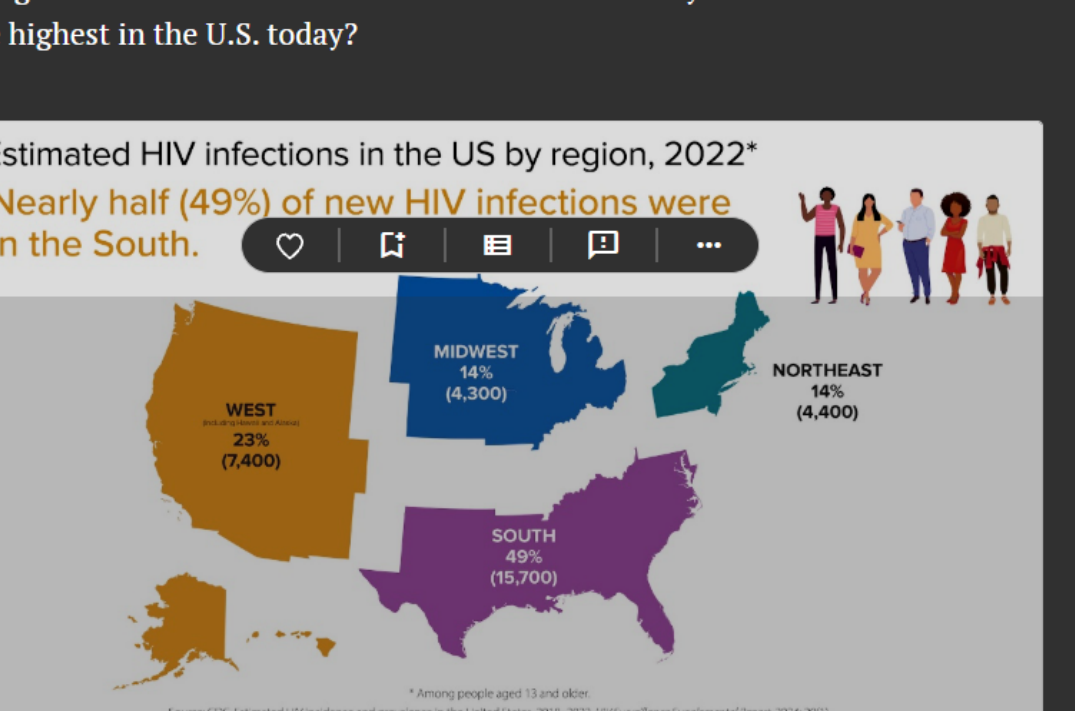

Estimated HIV infections in the US by region, 2022*
Nearly half (49%) of new HIV infections were in the South.
WEST
23%
(7,400)
MIDWEST
14%
(4,300)
NORTHEAST
14%
(4,400)
SOUTH
49%
(15,700)
* Among people aged 13 and older.

Figure 9

Credit: CDC

The South accounts for roughly half of all new cases of HIV and experiences the highest mortality rates compared to other regions. Structural barriers (poverty, stigma, lack of Medicaid expansion) and rural healthcare challenges (fewer prevention resources, limited health access to care) are key drivers of these disparities.

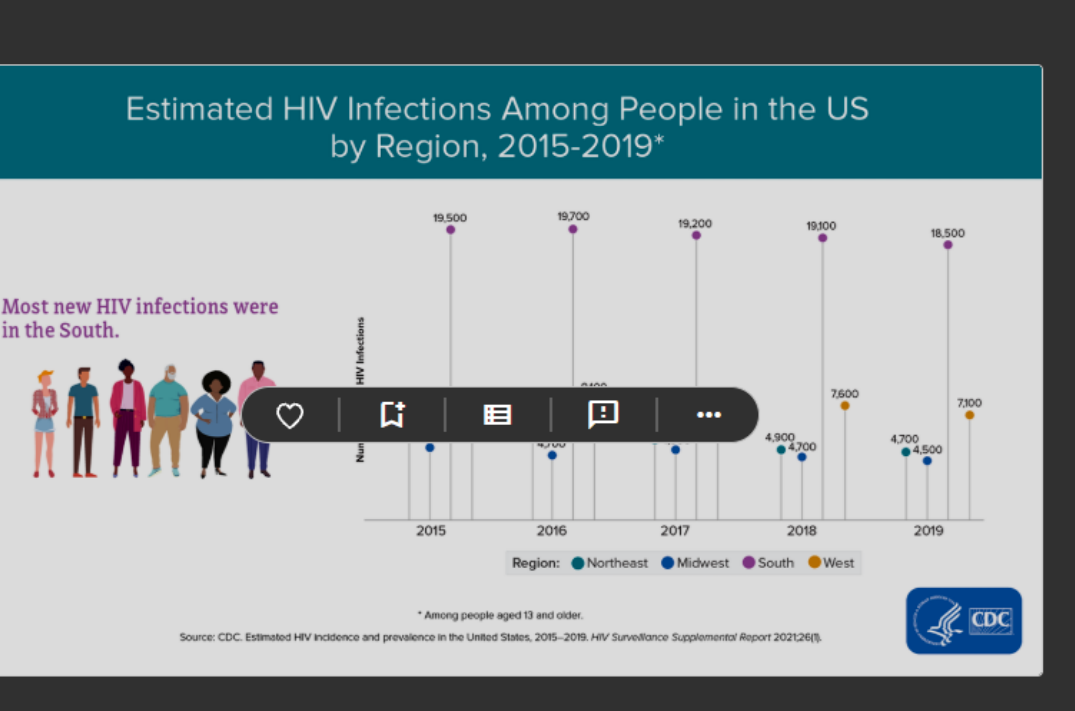


Figure 10

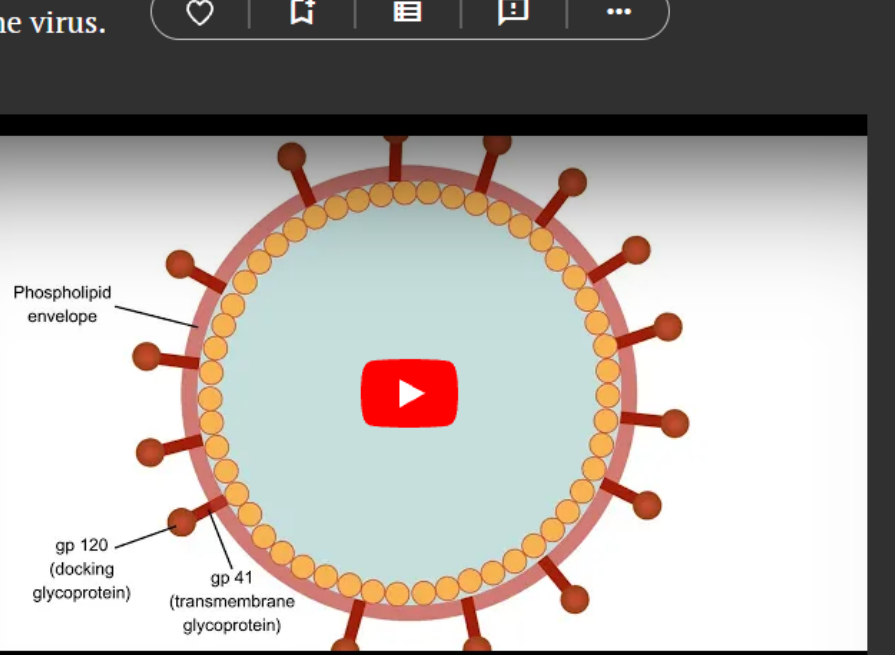
Only 87 out of 100 HIV+ patients in the U.S. were aware they had HIV. In the South, the number is substantially lower. Healthcare issues in the South continue to lead to disproportionate disease burdens not only for HIV, but for other diseases as well.

What's the structure of HIV? (LO3)

As mentioned previously, HIV has 9 genes total. Three of these genes encode major structural proteins and the enzymes HIV needs to complete its replication cycle.

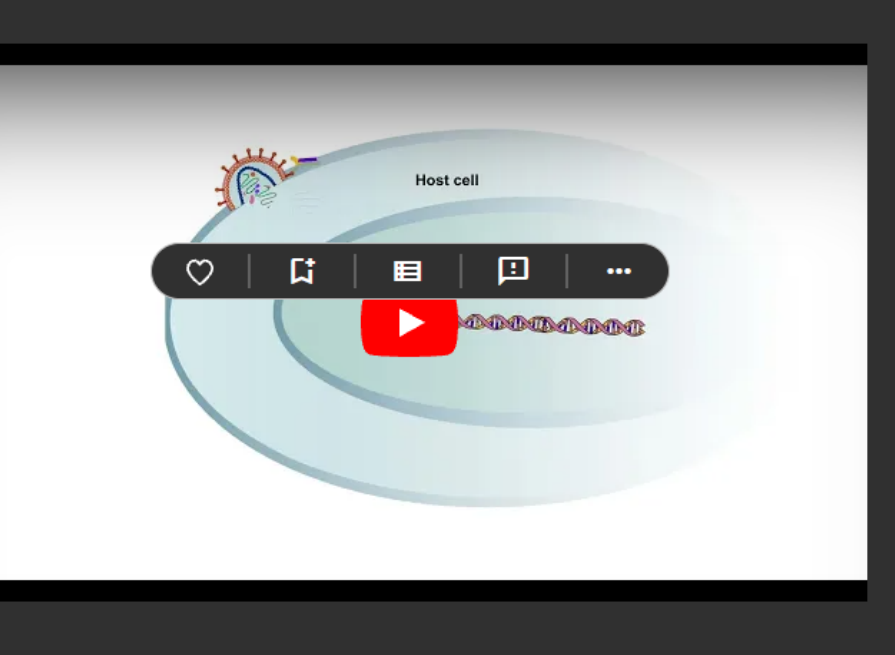
- **Gag gene (*gag*):** codes for the HIV structural proteins used in the capsid, nucleocapsid, and outer matrix (p24, p7, p17).
- **Env gene (*env*):** encodes gp120 and gp41, which mediate viral fusion with target cells.
- **Polymerase gene (*pol*):** encodes reverse transcriptase, integrase, and protease. All are critical for viral replication.

Knowing the structure of HIV is essential before we discuss the replication cycle of the virus.


Phospholipid envelope
gp 120 (docking glycoprotein)
gp 41 (transmembrane glycoprotein)

How does HIV replicate? (LO3)

Now we'll discuss the replication cycle of HIV. Note that many of these steps are targeted by ART at specific steps. You will need to know the steps of HIV replication and correlate them with your pharmacology brick discussing ART.



What cell types does HIV target and how does the virus enter into target cells?

In the next brick, we will discuss the clinical progression of HIV, the two criteria for an AIDS diagnosis, and the lab test used to screen for HIV.

Summary

- Retroviridae includes the medically significant viruses HIV and HTLV-1, which encode their genetic information as ssRNA, are enveloped, and have reverse transcriptase to convert viral RNA into DNA.
- HIV is a bloodborne illness, transmitted through sexual contact, parenteral exposure, occupational exposure, and vertical transmission.
- MSM, particularly among Black and Hispanic/Latino individuals, account for the majority of new HIV infections, but significant numbers of new cases also occur among heterosexual people, people who inject drugs, and individuals in regions with limited healthcare access, such as the Southern states.
- HIV targets cells by binding CD4 and a coreceptor: CXCR4 or CCR5.
- CD4 T cells are the major cell type that HIV infects; however, other cell types can be infected like macrophages and dendritic cells.
- Reverse transcriptase converts HIV RNA into DNA, integrase incorporates viral DNA into the host genome, and protease cleaves the polyprotein transcribed from viral DNA.
- HIV is difficult to cure due to the high mutation rate caused by reverse transcriptase and the permanent incorporation of viral DNA into the host genome.

Review Questions

1. A 43-year-old, HIV+ patient is on ART and during a routine HIV viral titer screening, his HIV titers have risen 4-fold. This patient has always been compliant with therapy. A resistance culture is ordered, and the virus has gained resistance to reverse transcriptase. Which HIV gene has mutated?

A. ○ *gag*
B. ● *pol*
C. ○ *env*

Hide Explanation

B) is correct. The *pol* encodes for reverse transcriptase, integrase, and protease. A) Capsid and other structural proteins are encoded by *gag*. C) Both gp120 and gp41 are encoded by *env*.

2. Which of the following is a characteristic of HIV?

A. ○ Replicates in the cytoplasm
B. ● Replicates in the nucleus
C. ○ Stores genetic information as DNA
D. ○ Binds target cells via CD21

Hide Explanation

B) is correct. Retroviruses are odd compared to other RNA viruses. Most RNA viruses replicate in the cytoplasm. HIV replicates in the nucleus. A) is incorrect. C) HIV stores its genetic info DNA once inside host cells. D) HIV binds CD4, not CD21. EBV binds CD21 to enter cells.

How would you rate this Brick?

References

- HIV | HIV | CDC
- Global HIV & AIDS statistics — Fact sheet | UNAIDS

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate between the taxonomy of Retroviridae and other viral families.~~
02 ☑ ~~Explain disease progression and epidemiology of HIV.~~
03 ☑ ~~Describe the life cycle of HIV.~~

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Objective Confidence: 3/3